Module 53

Backup & Recovery/3: Recovery/2

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Module Recap

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- A proper mix and management of volatile, non-volatile and stable storage can guarantee recovery from failures and ensure Atomicity, Consistency and Durability
- Log-based recovery is efficient and effective

Module Objectives

- To understand Transactional Logging with Hot Backup

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- To focus on concurrent transactions and understand the recovery algorithms

Module Outline

- Transactional Logging

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- Recovery Algorithm

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- Real time systems like sensor and actuator data in embedded devices, satellite transmissions etc. also use Hot backup

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- Thus they play a vital role in database recovery

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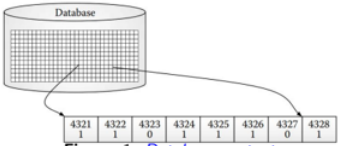

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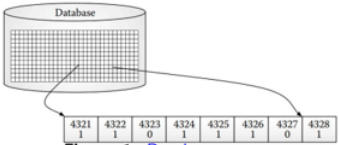

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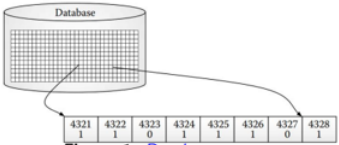

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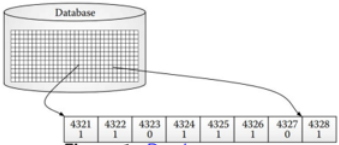

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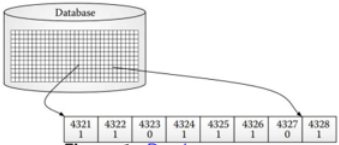

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¡5-¿ If a crash occurs before writing to the database then the inconsistent backed up file is recovered first, and then the pending modifications in the transaction log (backed up*) are applied to re-establish consistency

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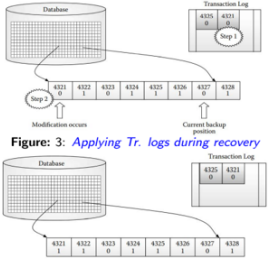

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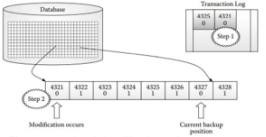

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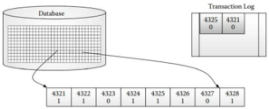

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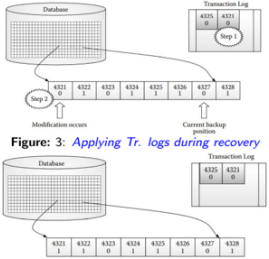

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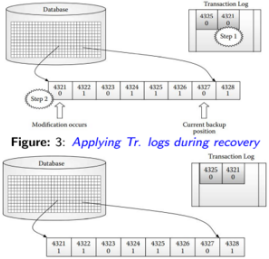

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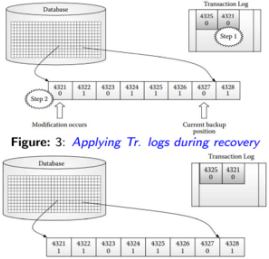

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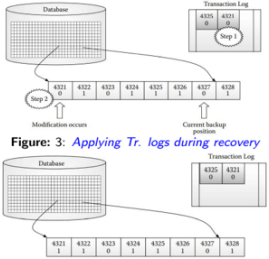

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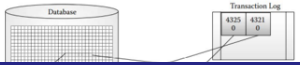

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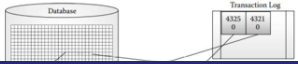
- Once all transaction logs have been replayed, the database is said to have been restored, that is, it is at a point where it can now be opened for user access



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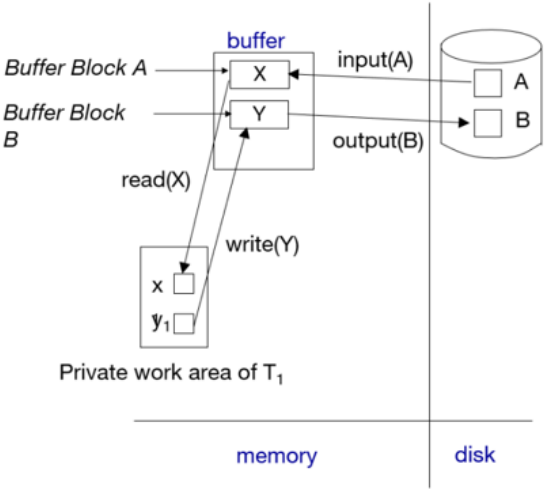
- Log records of different transactions may be interspersed in the log

Example of Data Access with Serial Transaction



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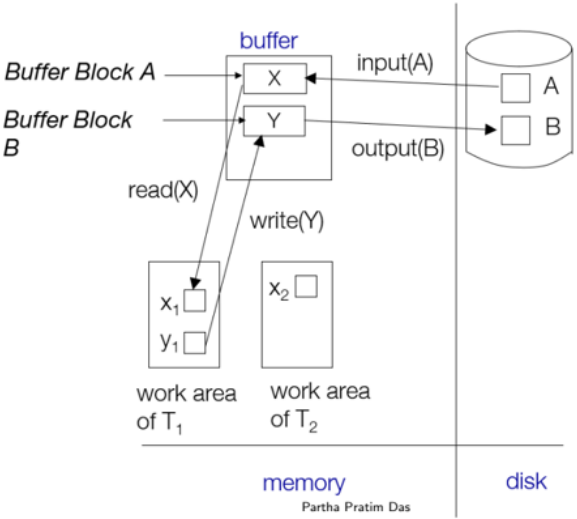

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Recovery Algorithm (3): Checkpoints Recap

- Let the time of checkpointing is t_{check} and the time of system crash is t_{fail}
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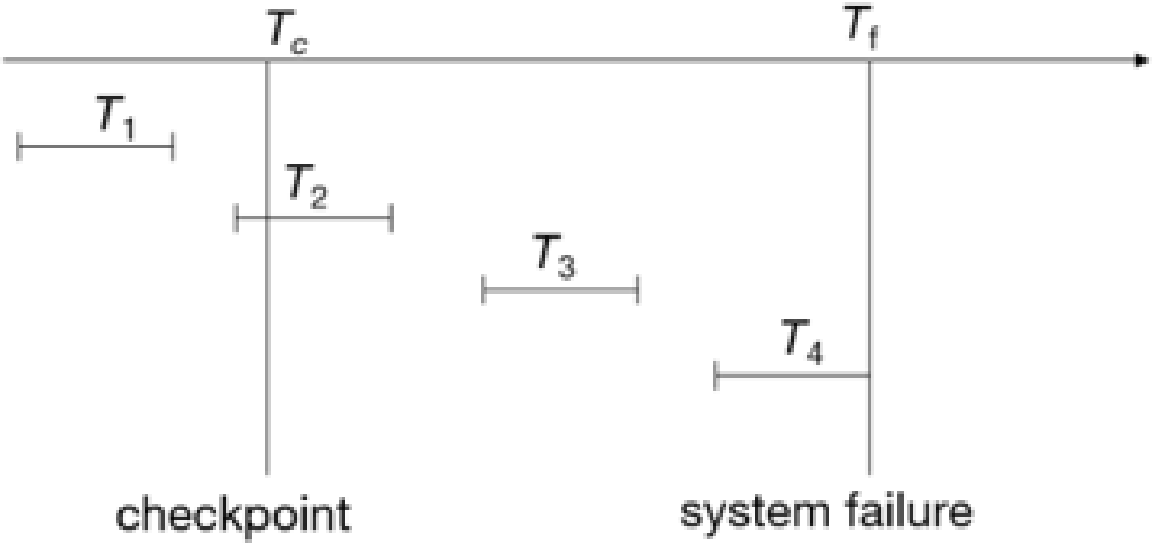
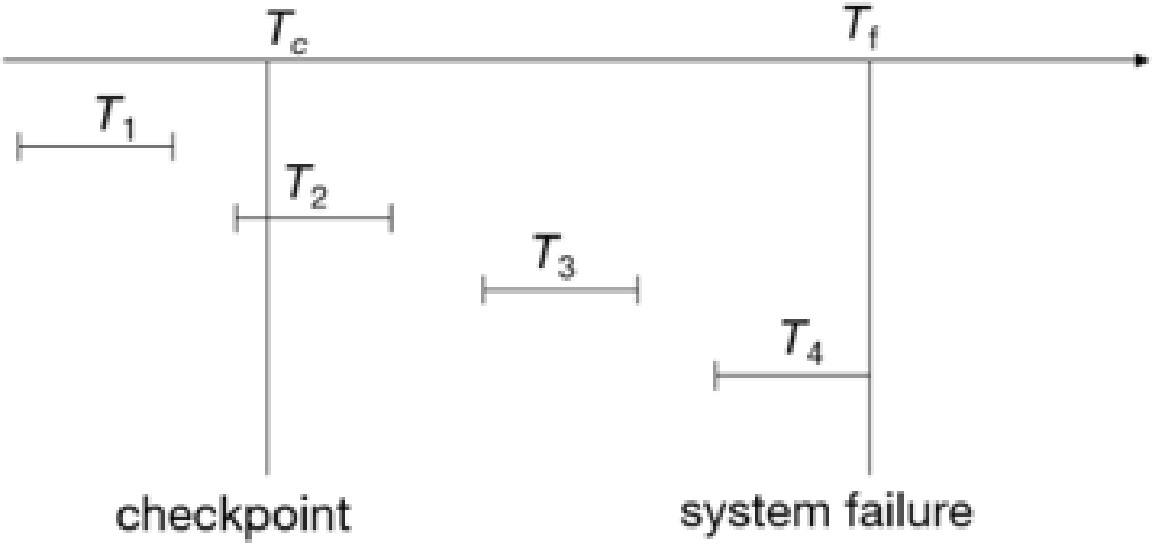
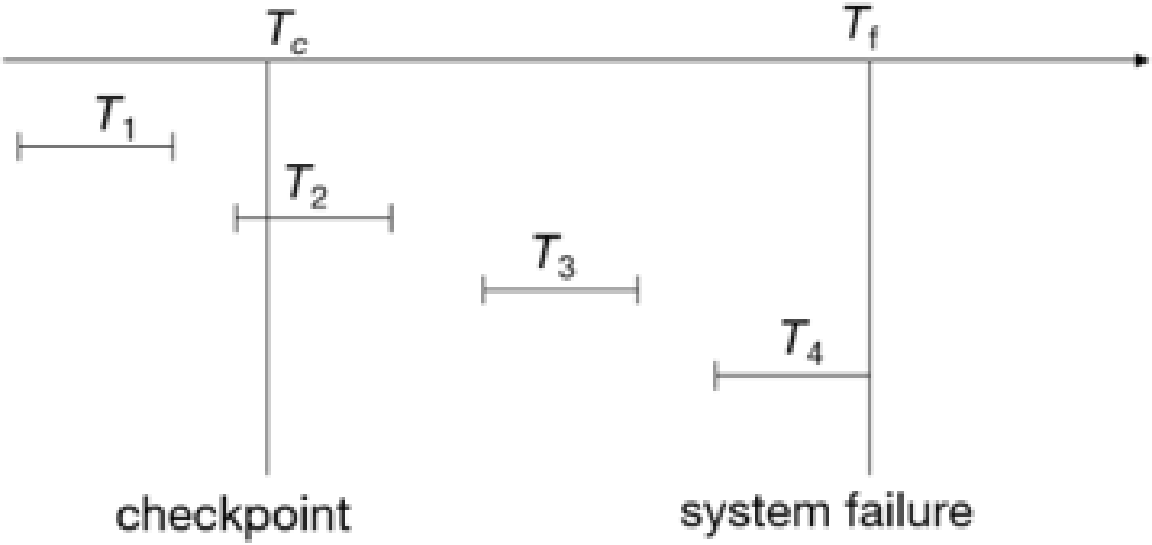
Recovery Algorithm (4): Checkpoints Recap



Recovery Algorithm (4): Checkpoints Recap



Recovery Algorithm (4): Checkpoints Recap



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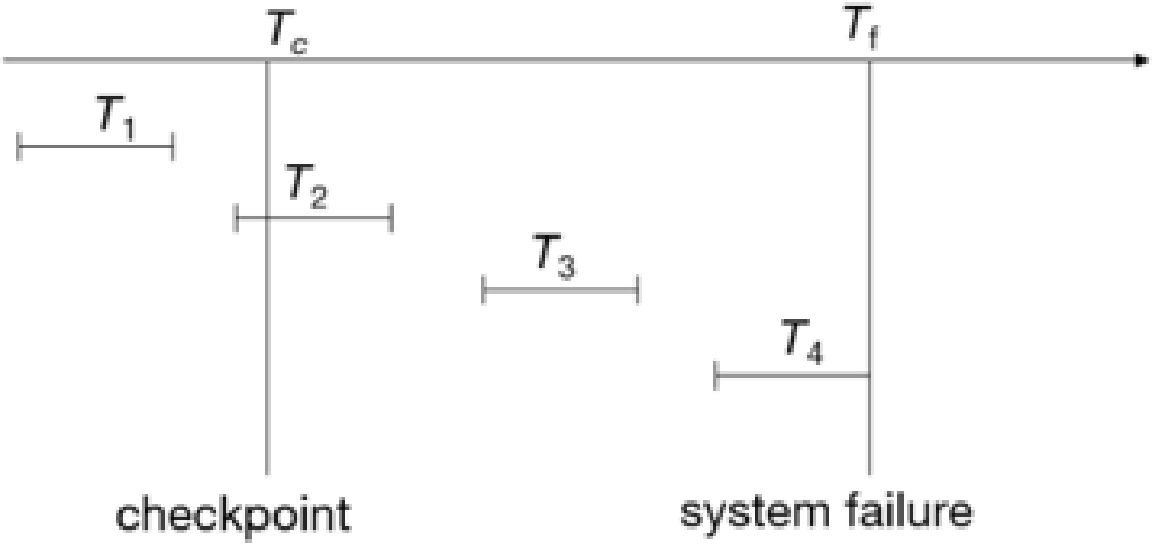

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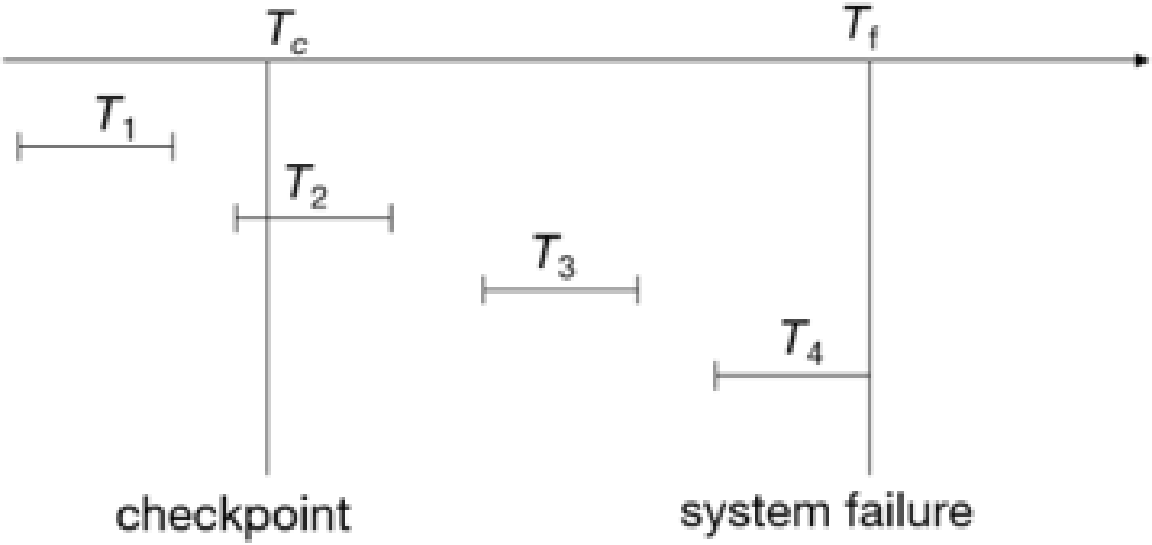

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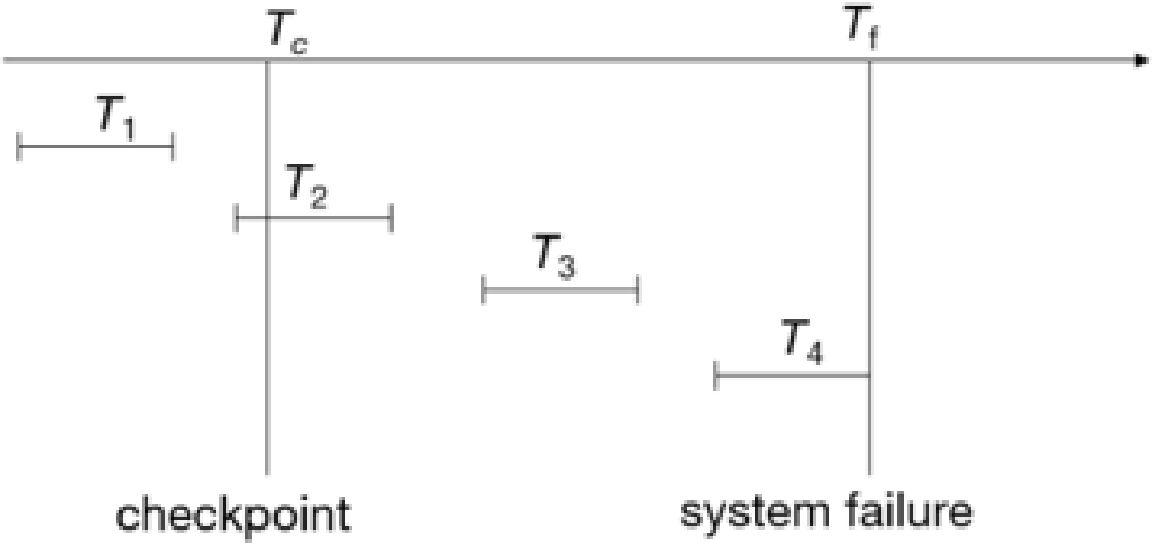

Recovery Algorithm (5): Redo-Undo Phases

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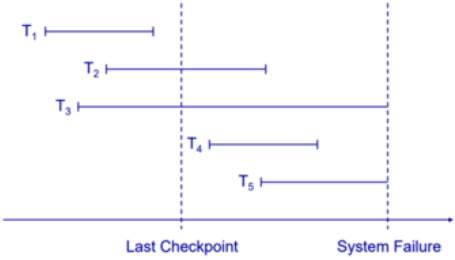

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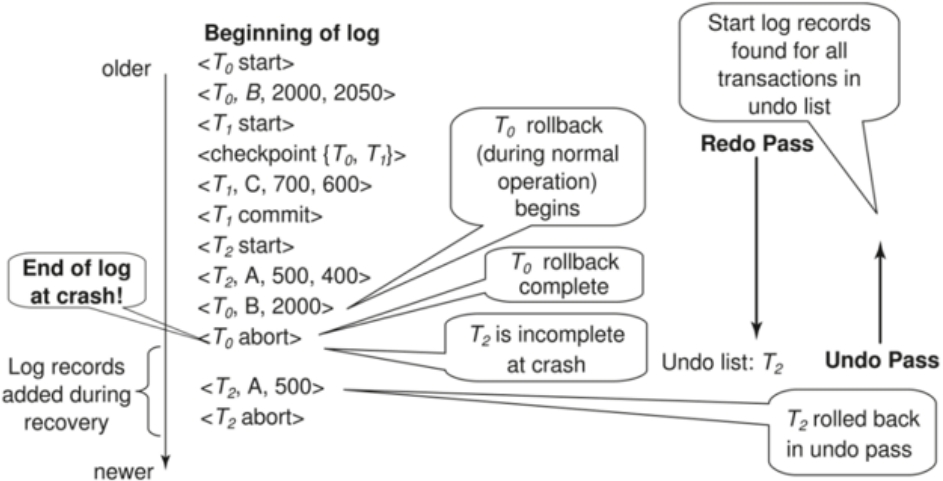
- After undo phase completes, normal transaction processing can commence

Recovery Algorithm (8): Example



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